Developer Handbook

Field	Value
Document ID	UAIL-ITDASH-DEV-001
Version	1.1
Status	Active baseline
Classification	Internal
Owner	Tech-Unit IT
Last Updated	2026-07-19
Audience	Developers, maintainers, deployment engineers

1. Purpose

Provide the working handbook for developers maintaining the UAIL IT Dashboard, including repository structure, local workflows, architecture touchpoints, coding rules, documentation generation, and validation steps.

2. Repository Structure

Path	Purpose
`api-gateway/`	central runtime state, admin APIs, WebSocket feed
`collectors/nutanix/`	Nutanix collector
`collectors/solarwinds/`	SolarWinds collector for servers and networks
`collectors/symphony/`	Symphony HSD collector
`dashboard/`	shared Next.js operator/admin UI
`frontdoor-proxy/`	front-door web processes
`deployment/`	staging, offline bundle, installer, support scripts
`docs/`	controlled project documentation

3. Workspace Model

The project uses npm workspaces from the root package.

Main workspaces:

- `api-gateway`
- `collectors/nutanix`
- `collectors/solarwinds`
- `collectors/symphony`
- `dashboard`
- `frontdoor-proxy`

4. Local Development Commands

Build all workspaces:

```
npm run build
```

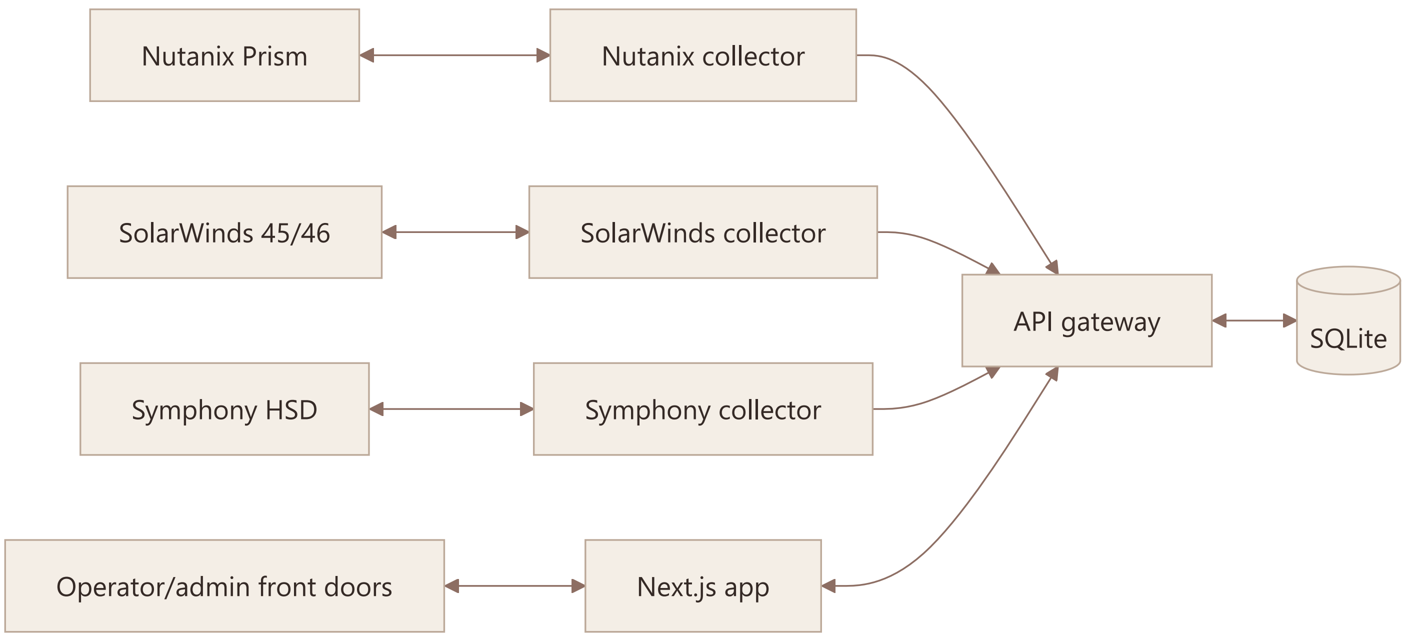
Start the full stack with PM2:

```
pm2 start ecosystem.config.js
```

Save the PM2 process list after validation:

```
pm2 save
```

5. Runtime Design Summary



Key rule:

- the gateway is the operational source for merged dashboard state

6. Source Rules Developers Must Preserve

- never introduce fabricated fallback values for normal operation
- preserve last-success state and freshness when collection fails
- keep Nutanix authoritative for HCI-backed servers until explicit fallback conditions are met
- keep SolarWinds authoritative for network telemetry
- treat session failure as a real source failure, not as a cosmetic condition

7. UI Development Rules

- protect the `1920x1200` wallboard layout first
- preserve responsive fallback behavior for tablet and mobile
- prefer graphical density over unnecessary text
- keep card-level stale/error messaging concise
- do not add visual features that hide data-source or freshness truth

8. Collector Maintenance Notes

8.1 Nutanix

- direct API collector
- sensitive to source certificate constraints
- primary source for HCI-backed servers

8.2 SolarWinds

- Playwright-based
- session-state dependent
- handles both server and network telemetry
- preserve separate credential handling for `SW_SERVERS_*` and `SW_NETWORKS_*`

8.3 Symphony HSD

- Playwright-based
- session-state dependent
- supports reauthentication and explicit legacy-profile import
- uses its own `SYM_*` credentials and session state, independent from SolarWinds

9. Admin Surface Responsibilities

The admin surface is not just a UI add-on. It is part of the operating model and must continue to support:

- service visibility and restart control
- session validation and recovery
- source configuration
- documentation access

10. Documentation Workflow

The retained controlled docs are:

- Product Requirements Document
- Project Documentation and Timeline
- System Design
- User Manual
- Developer Handbook

Regenerate PDFs with:

```
node docs/tools/export-markdown-pdfs.mjs
```

This writes PDFs to:

- `docs/pdf/`
- `dashboard/public/help/`

11. Deployment-Related Developer Notes

- deployment is SQLite-first
- PM2 is the current supervision model
- offline bundle generation remains important for Windows server deployment
- documentation changes that affect operations should also be reflected in the admin Help PDF set
- when bootstrap behavior changes, validate all three paths: source repo, offline bundle, and rebuilt installer artifact

12. Validation Checklist For Changes

12.1 Code Changes

- `npm run build`
- validate the affected workflow live where possible
- confirm no source-of-truth rule was broken

12.2 Documentation Changes

- regenerate PDFs
- verify Help tab references still match generated filenames
- check screenshot-heavy PDFs for print layout issues

12.3 Deployment Changes

- refresh staged payload when required
- validate the offline deployment path
- confirm the stack still boots and recovers under PM2

13. Common Pitfalls

- assuming a portal session file is valid without live validation
- masking stale data as current data
- changing deployment docs without updating admin Help artifacts
- tuning mobile layout in a way that regresses the primary wallboard layout

14. Maintainer Principle

The project should remain operationally honest before it becomes feature-rich. If a change improves visual appeal but weakens data trust, source traceability, or recovery clarity, it is the wrong change.